

Lionael Surya Dwitama

Toronto, ON | +1 437-606-8490 | suryalionael@gmail.com
[linkedin.com/in/lionael-dwitama](https://www.linkedin.com/in/lionael-dwitama) | github.com/suryalionael | lionael-portfolio.vercel.app

PROFESSIONAL SUMMARY

Data Science and Analytics student who builds backend and data systems end to end: a Kafka, Spark, and Delta Lake streaming pipeline with effectively exactly-once writes, idempotent PostgreSQL ETL workflows, and a Next.js, TypeScript, and Supabase app used daily in production. Shipped Python payroll-tax software as an intern, where correctness is not optional.

EDUCATION

Seneca Polytechnic

Honours Bachelor of Data Science and Analytics

Toronto, ON

Expected Apr 2028

- Cumulative GPA 3.9/4.0; President's Honour List (Winter 2026); SSF International Student Merit Award
- Coursework:** Advanced Database Design, Structured Database Design, Algorithms and Data, High Performance Computing

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Backend & Data: PostgreSQL, Supabase, Kafka, Spark Structured Streaming, Delta Lake, Airflow, dbt, DuckDB, FastAPI

Web: Next.js, React, Tailwind CSS

DevOps & Testing: Docker, Docker Compose, Git, GitHub Actions (CI), pytest, Playwright, Ruff, mypy

ML & AI: XGBoost, scikit-learn, SHAP, Claude API

Concepts: idempotent processing, ACID transactions, star-schema modelling, event-driven architecture, automated testing

PROJECTS

Streaming Clickstream Pipeline

Personal project

Python, Kafka, Spark Structured Streaming, Delta Lake, MinIO, DuckDB, Docker, GitHub Actions

- Built a real-time pipeline that streams about 50 events/sec from a Python producer into a 6-partition Kafka topic and processes them with Spark Structured Streaming using event-time processing and watermarking for late data.
- Designed a Bronze/Silver/Gold layer on Delta Lake (ACID) with validation and deduplication; achieved effectively exactly-once writes through idempotent foreachBatch commits and checkpointing, and routed unparseable messages to a dead-letter table.
- Orchestrated 7 services in Docker Compose with health checks, added a live dashboard with pipeline-health monitoring, and wrote 74 tests (unit, edge-case, Spark integration, Docker smoke) enforced in CI with Ruff and strict mypy.

Retail Sales ETL Pipeline

Personal project

Python, Apache Airflow, PostgreSQL

- Built a batch ETL pipeline that validates a real retail dataset and loads a star-schema PostgreSQL warehouse via Airflow.
- Designed the load step so reruns are safe by design and do not duplicate or corrupt warehouse data.

Aspen OS

In production at QuickRN

Next.js, TypeScript, Supabase (PostgreSQL), Tailwind CSS, Vercel, Playwright, GitHub Actions

- Built and deployed, as sole developer, a project operating system (tasks, kanban, docs) that a nonprofit team uses daily.
- Managed the PostgreSQL schema through versioned Supabase migrations, wrote Playwright end-to-end tests, and documented performance investigations of data fetching, production runtime, and bundle size.

IDX Stock Scanner

Personal project

Python, XGBoost, Claude API, yfinance, Parquet

- Built a daily decision-support agent for Indonesian equities with a three-layer design: a rules filter screens candidates, an XGBoost model ranks them, and an LLM explains each result rather than deciding it.

EXPERIENCE

Web Development & Data Operations, QuickRN (Remote)

Jul 2025 – Present

- Delivered three production websites from concept to deployment and built Aspen OS, the internal platform the team uses daily.
- Resolved website and online-course access issues through troubleshooting, managed exam-prep content data, and documented technical issues and fixes for customer inquiries.

Software Engineering Intern, CV. Veda Sakti Dharma, Jakarta, Indonesia

Jun 2025 – Aug 2025

- Developed and tested a Python app that automates PPH 21 (Indonesian income tax) calculations and employee paylips.
- Debugged software and validated payroll data to improve reliability and reporting accuracy; collaborated on new features.

IT & Media Strategy Associate, PERMIKA Toronto

Aug 2025 – Sep 2026

- Built and maintained internal tools for event management and member engagement; led the social media campaign for League of Toronto 2026, coordinating a team of ten.